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Magnetic overprinting of the Brachina Formation/Ulupa Siltstone, Southern Adelaide Foldbelt, prior to Delamerian deformation

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The Proterozoic sediments of the Adelaide Geosyncline were deformed and metamorphosed during the Ordovician Delamerian Orogeny. In much of the Flinders Ranges, the relatively unmetamorphosed sediments, such as those of the Brachina and Bunyeroo formations in the Central Flinders Ranges, are only weakly magnetic. In the Mount Lofty Ranges, where peak metamorphic grades were reached, the Brachina Formation and its equivalent, the Ulupa Siltstone, are extremely magnetic and cause prominent linear magnetic anomalies with amplitudes of 100 to 500 nT that can be traced for more than 50 km. The higher metamorphic grades are accompanied by the growth of metamorphic magnetite within beds that are thought to have originally contained siderite. These magnetic markers have been used to trace the macrostructure of the Southern Adelaide Foldbelt and have been especially useful where outcrop is limited. At the highest grade, where the rocks are andalusite schists, the magnetic anomalies are inconsistent with magnetisations in the direction of the present magnetic field implying that remanence causes a significant proportion of the anomaly. The rocks display a strong thermochemical component, mostly acquired prior to deformation due to the Delamerian Orogeny. Remagnetisation prior to deformation is consistent with the high-temperature low-pressure metamorphism that characterises the eastern Mt Lofty Ranges. This secondary natural remanent magnetisation (NRM) component is similar to that of the nearby post-orogenic Black Hill Norite, which is believed to have intruded during the waning phases of the Delamerian Orogeny and is of the same magnetic polarity. We therefore have magnetic signatures of the same polarity that bracket the Delamerian Orogeny, both pre- and post-folding. The tilt corrected mean direction for the Ulupa Siltstone from 44 samples is $D = 231.4°$, $I = 41.1°$ ($\alpha_{95} = 5.6°$), and a pole position at latitude $\lambda_p = 46.0°$S, longitude $\phi_p = 49.6°$E ($A_{95} = 6.4°$).

KEY WORDS: South Australia, Neoproterozoic, paleomagnetism, aeromagnetic, Delamerian, polar wander

INTRODUCTION

It is widely appreciated that magnetic anomalies are important adjuncts to geological mapping but their potential for constraining aspects of geological history is underestimated. Rock magnetism combined with conventional petrology can identify the processes that create, alter and destroy magnetic minerals in rocks. By relating magnetic mineralogy, bulk magnetic properties, petrology and geochemistry to observed magnetic anomalies, an understanding of the geological factors that control magnetic signatures is obtained. This can be used to improve geological interpretation of magnetic surveys.

Many magnetic anomalies throughout the Adelaide Foldbelt have long been known to be predominantly caused by magnetic remanence, rather than induced magnetisation (Tucker 1972; McKirdy *et al.* 1975). These anomalies are the magnetic signatures of metamorphism associated with the Delamerian Orogeny. Peak metamorphic grades were reached in the Southern Adelaide Foldbelt (in the Mount Lofty Ranges east of Adelaide), where the Brachina Formation and its equivalent, the Ulupa Siltstone, are extremely magnetic. Folds and other structures of the foldbelt can be traced using the magnetic markers within these formations, which is especially useful where outcrop is limited. While the magnetic markers are consistent with respect to their continuity and presence, the amplitude and character of the anomaly changes significantly along strike, being a function of metamorphic grade. Anomalies from lower grade areas can be successfully interpreted assuming induced magnetisation only. In the highest-grade areas, where the rocks are andalusite schists, the magnetic anomalies are inconsistent with known structures and the direction of the local magnetic field, and remanence must be invoked to explain the majority of the anomaly.

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SOUTHERN ADELAIDEAN STRATIGRAPHY AND GEOCHRONOLOGY

A geological and locality map is shown in Figure 1. A stratigraphic column for the Cryogenian and Ediacaran strata of the southern Flinders Ranges is given in Figure 2, following Preiss (1990) and Preiss *et al.* (1993). The Ulupa Siltstone is a deeper water more distal equivalent of the Brachina Formation, and immediately overlies the Nuccaleena Formation at the base of the Ediacaran Period. The Ulupa Siltstone appears to be more reduced than the Brachina Formation, being green-grey to mauve rather than chocolate brown to red in colour.

Ages for the the Neoproterozoic Cryogenian–Ediacaran succession of the Adelaide Foldbelt are not definitive and tend to be indirect, either Rb–Sr whole-rock shale isochrons or other radiometric ages for scarce igneous units or detrital grains. The 827 ± 6 Ma Gairdner Dykes (Wingate *et al.* 1998), widely accepted as feeders for the basal-Adelaidean Beda Volcanics (Mason *et al.* 1978), intrude the pre-Adelaidean Pandurra Formation (Schmidt & Williams 2011) but not the Tapley Hill Formation. A Re–Os age of 643 ± 2.4 Ma for the Tindelpina Shale member of the Tapley Hill Formation (Kendall *et al.* 2006) and a zircon age of 657 ± 17 Ma for Wilmington Formation (Ireland *et al.* 1998) are indicative ages for the interglacial period. Mahan *et al.* (2010) reported a Th–U–total Pb age of 680 ± 23 Ma for authigenic monazite from sandstone from the Enorama Shale. This diagenetic age for the Enorama Shale is compatible within errors with the age of 657 ± 17 Ma for the Wilmington, and even the ~659 Ma age for the Wilyerpa Formation below the Tapley Hill Shale, but it conflicts with the age of 643 ± 2.4 Ma for the Tindelpina Shale Member.

The age of the Elatina Formation at the end of the Cryogenian, is inferred to be 635 Ma on the basis of global correlation but Kendall *et al.* (2006) noted that their age of 643 Ma for the Tindelpina Shale requires excessive rates of sedimentation for the >4 km interglacial succession in the central Flinders Ranges. Based

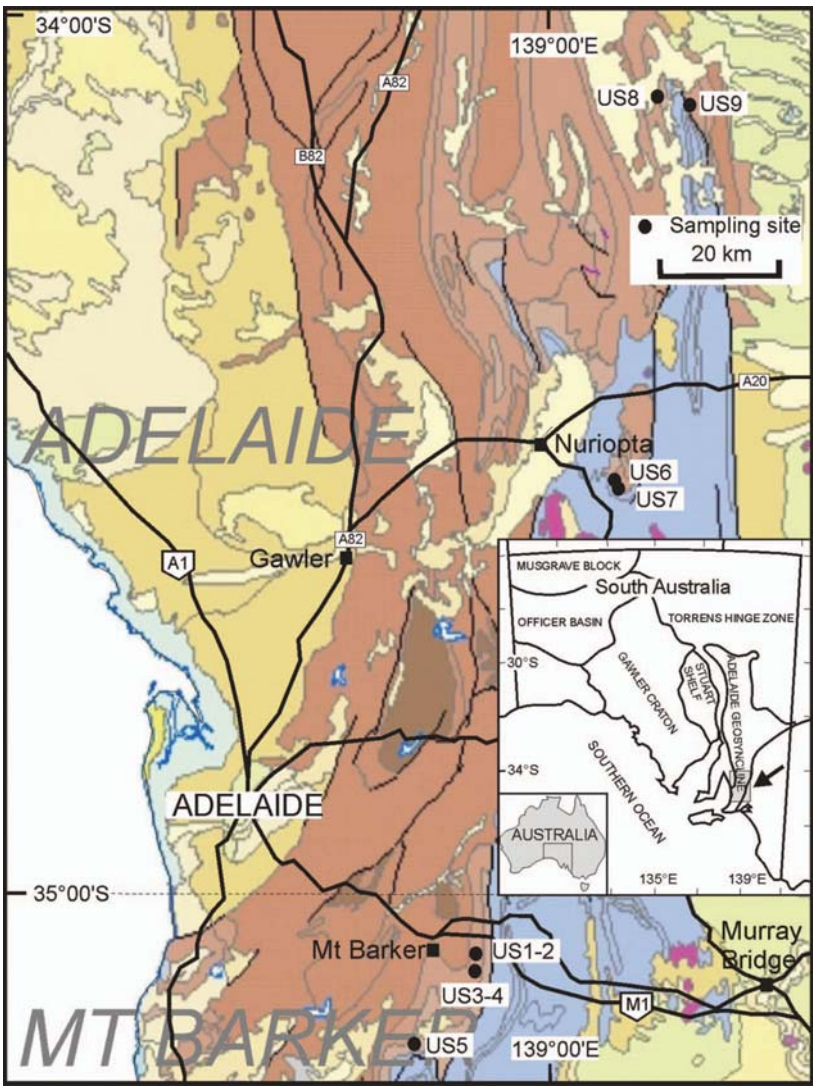


Figure 1 Geological map of the southern and southeastern Flinders Ranges. Site details are given in Appendix 1. Background adapted from Surface Geology of Australia 1:1 000 000 scale, South Australia © Commonwealth of Australia (Geoscience Australia) 2010, URL: <http://www.ga.gov.au/ausgeonews/ausgeonews200809/productnews.jsp#product1> (released under the Creative Commons Attribution 2.5 Australia Licence).

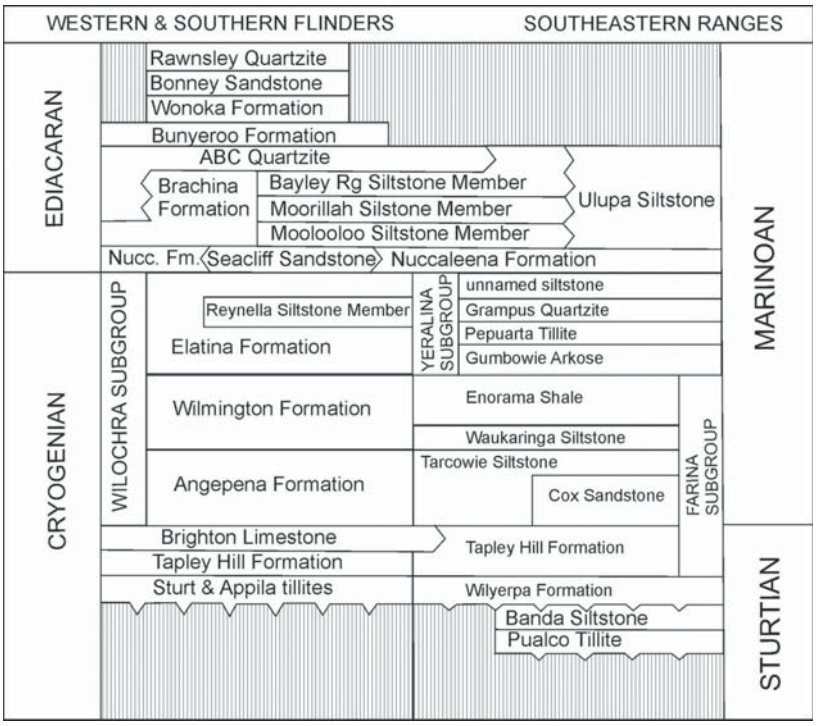


Figure 2 Stratigraphic column for the western, southern and south-eastern Adelaidean.

on correlation with Tasmanian sequences it has been argued that the Elatina Formation is more likely 580 Ma (Calver *et al.* 2004). This would alleviate the problem of excessive rates of sedimentation but the age of the Elatina Formation remains somewhat controversial at present.

For the Ediacaran units, Compston *et al.* (1987) obtained Rb–Sr isochrons of 609 $\pm$ 64 Ma for the Brachina Formation and of 593 $\pm$ 32 Ma for the Yarloo Shale on the Stuart Shelf. The Yarloo Shale is the equivalent to the Bunyeroo Formation to the east, which overlies the Brachina Formation. The youngest of 11 ^{40}Ar–^{39}Ar ages from detrital muscovites from the Bonney Sandstone was found to be 601 $\pm$ 17 Ma (Haines *et al.* 2004) while a single zircon grain from the Bonney Sandstone yielded a U–Pb age of 556 $\pm$ 24 Ma (Ireland *et al.* 1998). This possibly records penecontemporaneous volcanism and provides a maximum age for the overlying Rawnsley Quartzite (Preiss 2000). The best estimate for the age of the Brachina Formation, and by inference, the Ulupa Silstone, is somewhere around 590 Ma.

The Adelaide Foldbelt sediments were deformed and metamorphosed during the Cambro-Ordovician Delamerian Orogeny. The metamorphism in the Mt Lofty Ranges was of short duration, high-temperature and low-pressure, apparently resulting from the ascent of hot magma into the crust (Sandiford *et al.* 1995). Two magma types have been recognised, I-types with zircon ages, believed to be crystallisation ages, of 515–490 Ma, and a younger bi-modal suite, dominantly of A-type affinity, that are undeformed and yield zircon ages of 485 Ma. The Black Hill Norite is related to this younger suite of intrusives.

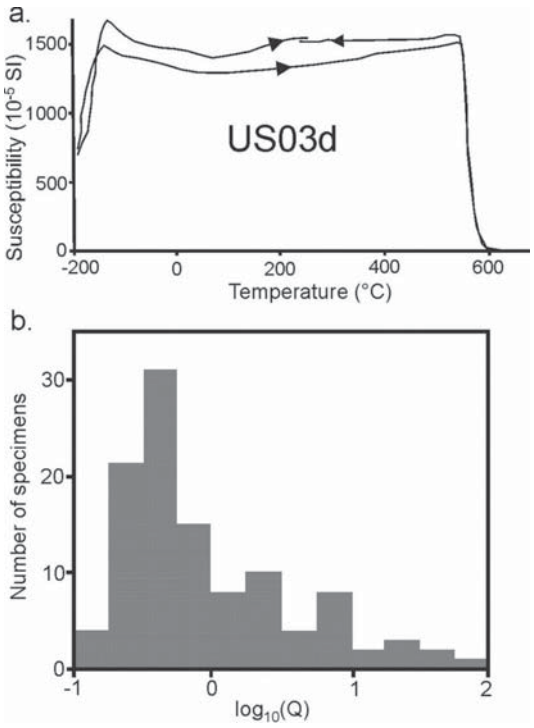


Figure 3 (a) Susceptibility *vs.* temperature curve identifying magnetite (the sample has been re-heated from LN_2 temperature after heating to above 600°C to check that the Verwey transistion is repeatable, as it is here, indicating that the magnetite is substantially unaltered on heating), and (b) $\log_{10}$ Koenigberger ratio histogram.

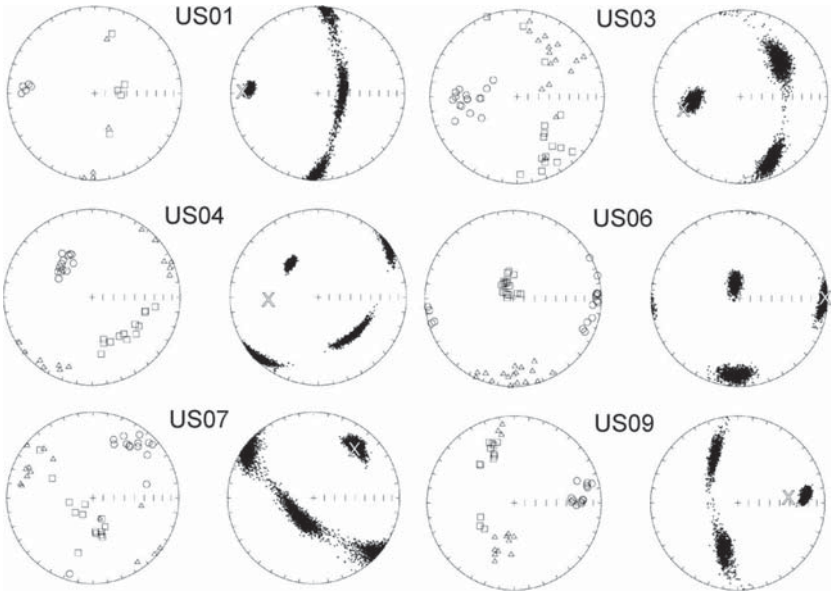


Figure 4 Stereographic projections of axes from anisotropy of magnetic susceptibility (AMS). On the left hand side, squares (circles, triangles) represent axes of maximum (minimum, intermediate) suceptibilities. On the right hand side, the data have been bootstrapped using the method of Tauxe (1998). The bedding planes are evident especially when the maximum and intermediate susceptibilities are approximately equal and the fabrics are nearly oblate.

Table 1 Paleomagnetic summary for the Ulupa Siltstone (mean site $\lambda_s = 34.5°$S, $\phi_s = 139.0°$E).

Site	N	D_h (°)	I_h (°)	D_b (°)	I_b (°)	k	α_{95} (°)
US01	2	233.2	−32.6	234.1	30.8	8.24	—
US03	15	245.2	−31.0	244.1	35.6	14.9	4.0
US04	11	227.8	−2.4	210.8	45.3	46.9	6.7
US06	6	114.9	55.1	252.9	35.1	51.0	9.5
US07	5	229.6	33.5	229.6	33.5	820	2.7
US09	5	98.6	32.5	190.8	54.8	8.55	27.8
Sites	6	211.1	15.9	229.7	40.8	19.1	15.7
Specimens	44	223.3	2.1	231.4	41.1	15.9	5.6

Structurally corrected pole position based on sites and specimen VGPs: $N = 6$, kappa = 5.7, $\lambda_p = 46.9°$S, $\phi_p = 48.0°$E, $A_{95} = 18.3°$, $N = 44$, kappa = 12.3, $\lambda_p = 46.0°$S, $\phi_p = 49.6°$E, $A_{95} = 6.4°$. N, number of specimens/sites showing ChRM; Dh and Ih, *in situ* declination and inclination; Db and Ib, declination and inclination with respect to bedding; k, precision parameter kappa, α_{95}/A_{95}, half-angle of the 95% confidence cone for the mean direction/VGP (Fisher 1953). Site locations used for VGP calculations are given in Appendix 1.

TECHNIQUES

Samples were collected using a power drill and oriented both with magnetic and sun compasses. The magnetic declination of the local field was found to be consistent with the International Geomagnetic Reference Field data. Between three and nine specimens 2.2 cm high × 2.5 cm in diameter were machined from the cores. Routine paleomagnetic laboratory methods (Collinson 1983; Butler 1992) were employed.

Susceptibility and its variation with temperature were measured using an in-house transformer bridge calibrated using a paramagnetic salt. Anisotropy of magnetic susceptibility (AMS) was measured using a DIGICO anisotropy delineator interfaced to a personal computer (PC). Remanent magnetisation was measured using a CTF three-axis cryogenic magnetometer interfaced to a PC. All specimens were subjected to stepwise thermal and/or AF demagnetisation with 10–15 steps. The demagnetisers used were an in-house carousel furnace housed within a 10-coil Helmholtz set with feed-back to keep the field within ±1 nT on all components, and a Schoenstedt AF demagnetiser. Magnetisation components were isolated using an interactive version of Linefind (Kent *et al.* 1983), in which linear segments are fitted to data points weighted according to the inverse of their measured variances.

SAMPLING SITES

The locations of sampling sites are given in Appendix 1. Sampling was guided by reference to airborne magnetic maps and the use of a handheld Elliot susceptibility metre. Generally areas selected for sampling displayed susceptibilities of at least 200–300 × 10^{-5} SI. Attention to geological structures ensured representative strata

with a spread of geological strikes and dips were sampled (see Figure 1).

ROCK MAGNETISM

A susceptibility versus temperature (k–T) curve for sample US03d (Figure 3a) from site 03 reveals a maximum Curie temperature of 580°C, indicating that magnetite is the main magnetic mineral, consistent with the intense remanence noted above. The low temperature peak near –150°C also indicates magnetite. This is the Verwey transition where magnetite undergoes a crystallographic transition, briefly allowing magnetisation to align with the applied field increasing the observed susceptibility. The k–T curve is almost reversible, a characteristic of pure magnetite that contains only Fe cations. Such pure magnetite, while often occurring in magmatic rocks, when found in abundance in sediments is typically caused by thermal metamorphism, often as a result of the oxidation of siderite.

Many sites showing a strong magnetic susceptibility in the field also possessed a strong magnetic remanence (as measured in the laboratory) but for some sites with significant susceptibilities, this did not translate into significant remanent intensities. Thus, Koenigsberger ratios for the Ulupa Siltstone varied from quite low to very high (Figure 3b). Some 10% of values are in excess of 10 and several from sites 03 and 06 were even higher, 20–80. These values are extreme and normally would suggest lightning as the cause. However, the natural remanent magnetisation (NRM) directions from sites 03 and 06, in particular, show a high degree of consistency indicating that the NRMs genuinely reflect the magnetisation of Ulupa Siltstone strata that carry an intense remanence with intensity >10 Am^{-1}.

Axes of anisotropy of magnetic susceptibility for specimens from each site which yielded useful data are plotted in Figure 4. For each site the measured axes are plotted on the left while boot-strapped axes (Tauxe 1998) are plotted on the right giving a better sense of the bedding plane. The intermediate and maximum axes tend to be spread within the bedding plane, indicating oblate susceptibility ellipsoids with the minimum susceptibility axes close to the bedding poles. The bedding poles measured in the field (as determined from the structures listed in Appendix 1) generally agree well with the mean minimum susceptibilty axes, except for two sites. Site US04 shows a 30° departure of the field observation from the AMS data. It is possible that this site shows some strain effects although this has not been pursued further. Although the susceptibility is dominated by the metamorphic magnetite, the data show overwelmingly that the AMS is a mimetic fabric reflecting the original bedding.

PALEOMAGNETISM OF THE ULUPA SILTSTONE

Demagnetisation

The results are summarised in Table 1, while Figure 5 shows orthogonal projections of NRM vectors and

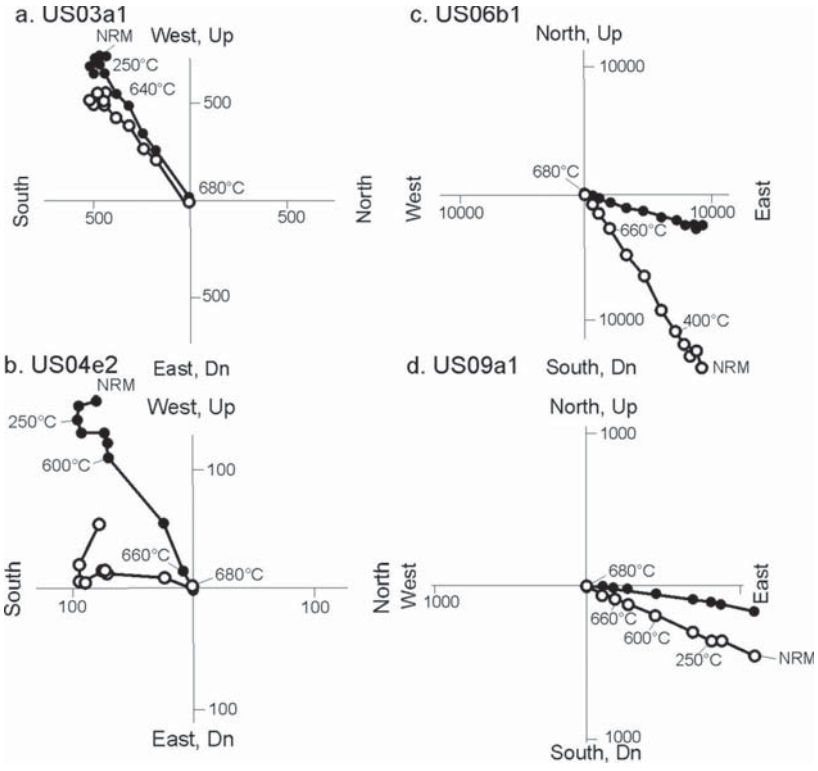


Figure 5 Orthogonal projections of remanence vectors for specimens from the Ulupa Formation after stepwise thermal demagnetisation. Solid circles plot on the horizontal plane and open circles on the vertical plane. Axes are mAm^{-1} (10–6 emu/cc), temperature in°C. See text for details.

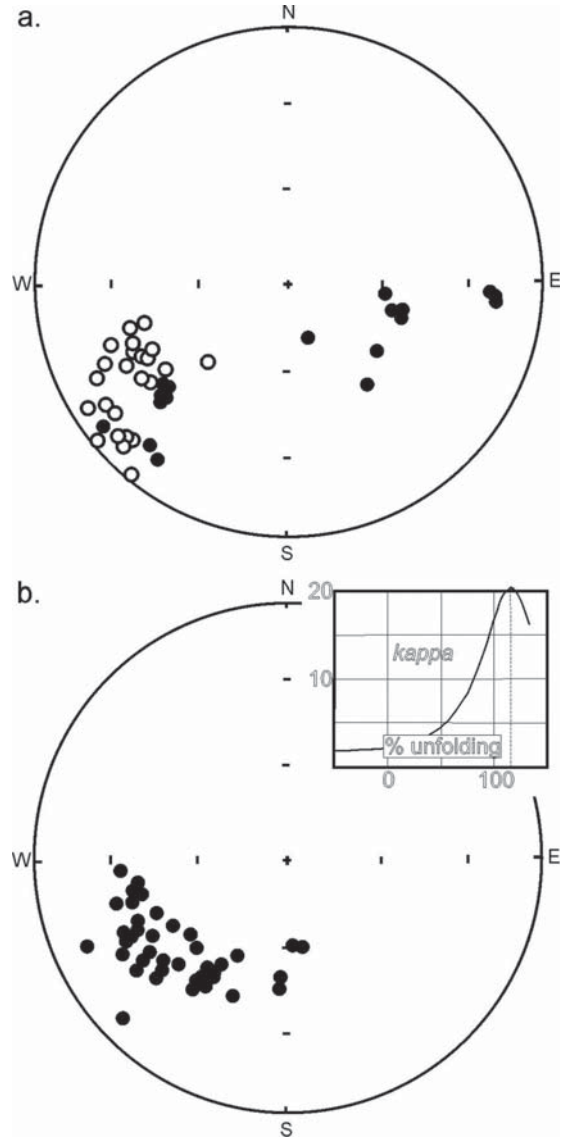


Figure 6 Stereographic projections of characteristic remanent magnetisation (ChRM) directions for the Ulupa Formation isolated by stepwise thermal demagnetisation. Solid circles plot on the lower hemisphere and open circles on the upper hemisphere: (a) present-day coordinates, and (b) after structural correction. Inset shows the variation of the precision parameter with degree of unfolding.

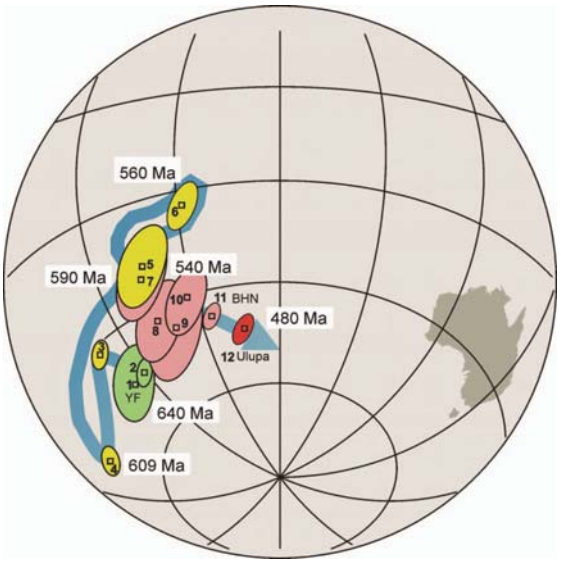


Figure 7 Apparent polar wander path for late Paleoproterozoic and through to the Mesoproterozoic. Cryogenian poles are green, Ediacaran are yellow, Cambrian are pink while the Ulupa Siltstone pole (herein) is red. Numbered poles after Schmidt *et al.* (2009) are 1, Yaltipena Fm; 2, Elatina Fm; 3, Nuccaleena Fm; 4, Brachina Fm; 5, Bunyeroo Fm; 6, Wonoka Fm; 7, Hawker Gp; 8, Billy Ck Fm; 9, Kangaroo Island redbeds; 10, Lake Frome Gp (basal); 11, Black Hill Norite.

sequential vectors after stepwise thermal demagnetisation up to 580°C. Generally after 580°C samples were completely demagnetised although the magnetisation of some samples, particularly from site 03, persisted to above 600°C. The intensity of magnetisation of this component appears to be too strong to ascribe to being carried by hematite, e.g. the intensity remaining above 660°C for specimen 6b (Figure 5c) is several Am^{-1}. While not ruling out the possibility that hematite is present, strong remanence remaining above 600°C is interpreted as resulting from incipient weathering, leading to leaching of Fe and a cation deficient magnetite crystallographic structure, i.e. partial maghemitisation. Maghemite, which is cation deficient, is known to have a Curie temperature above that of pure magnetite (Dunlop & Özdemir 1997, p. 398). The inversion temperature of maghemite to hematite has been reported as 250°C to >750°C, and naturally occurring maghemites have higher, and often much higher, inversion temperatures than synthetic maghemites (Dunlop & Özdemir 1997, p. 59), from which most estimates have been made. Therefore, magnetite (*s.l.*) bearing rock that has been subjected to weathering may display Curie temperatures above 600°C, simply reflectling partial maghemitisation.

There is a close correlation between metamorphic grade and the variation in magnetic mineralogy and natural remanent magnetisation. The rocks display multiple components of NRM, including a weak component in the direction of the present magnetic field and a much stronger thermo-chemical component acquired during the Delamerian Orogeny. After unfolding, this thermo-chemical component has a nearly identical direction as the stable magnetisation of the nearby Black Hill Norite (Schmidt *et al.* 1993), which has a radiometric age of 485 Ma and is believed to have intruded during the waning phases of the Delamerian Orogeny.

While it is clear that directions from most sites are brought into better agreement after untilting, some of the directions found at site US07 do not require untilting to conform with the untilted directions from other sites.

Table 2 Summary of reliable late Cryogenian to post-Delamerian Orogeny poles for late Neoproterozoic and early Paleozoic rocks, South Australia (quality factor Q at least 5/7).

Formation	Dec (°)	Inc (°)	α_{95} (°)	λ_p (°)	ϕ_p (°)	Dp (°)	Dm (°)	1	2	3	4	5	6	7	Q	Ref.
12 Ulupa Siltstone	231.4	41.1	5.6	−46.0	49.6	6.4†										This paper
11 Black Hill Norite	231.1	19.7	3.8	−37.5	34.4	3.0	6.0	√	√	√	√	√	×	√	6	a
10 Lake Frome (basal)	232.3	−0.5	10.1	−31.4	26.9	5.1	10.1	√	√	√	×	√	√	√	6	b
9 Kangaroo Is red beds	224.5	−4.4	12.3	−33.8	15.1	6.2	12.3	√	√	√	×	√	√	√	6	b
8 Billy Ck Fm	224.1	−1.2	14.4	−37.4	20.1	7.2	14.4	√	√	√	×	√	√	√	6	b
7 Hawker Group	233.4	−27.8	11.4	−21.3	14.9	6.8	12.5	√	√	√	×	√	√	√	6	b
6 Wonoka Fm	255.9	−23.7	6.4	−5.2	30.5	3.6	6.8	×	√	√	√	√	√	√	6	c
5 Bunyeroo Fm	236.6	−29.3	10.7	−18.1	16.3	6.5	11.8	×	√	√	√	√	√	×	5	d
4 Brachina Fm	178.2	−22.6	4.4	−46.0	315.4	2.4	4.6	×	√	√	√	√	√	√	6	c
3 Nuccaleena Fm	207.0	−36.0	3.7	−32.3	349.1	2.4	4.2	×	√	√	√	√	√	√	6	e
2 Elatina Fm	208.3	−12.9	4.2	−43.7	359.3	2.1	4.2	×	√	√	√	√	√	√	6	e
1 Yaltipena Fm	204.0	−16.4	11.0	−44.2	352.7	5.9	11.4	×	√	√	√	√	√	√	6	f

1–7 and Q, Reliability criteria of Van der Voo (1990). a, Schmidt *et al.* (1993); b, Klootwijk (1980); c, Schmidt & Williams (2010); d, Schmidt & Williams (1996); e, Schmidt *et al.* (2009); f, Sohl *et al.* (1999).
N, number of samples showing ChRM; α_{95}, half-angle of the 95% confidence cone for the mean direction; D_h and I_h, *in situ* declination and inclination; D_b and I_b, declination and inclination with respect to bedding; *k*, precision parameter of Fisher (1953) distribution for mean directions; λ_p, ϕ_p, latitude, longitude of the site mean pole dp, dm, semi-axes of error ellipse around the pole of probability of 95%.
†Single values represent A_{95}, half-angle of the 95% confidence cone for the mean pole position.

This implies that the magnetisation at site US07 post-dates the deformation. The remanent intensities of these samples were <10 Am^{-1}, which is one to two orders of magnitude less than at other sites and suggests that their mode of acquisition differs from the main group. It is possible that the remagnetistion process partially overlapped the deformation, but at most other sites the timing is distinctly different.

Directions before and after untilting are shown in Figure 6 (site US07 directions are uncorrected in both cases; see Table 1). A fold test (McFadden 1990) is positive with the correlation statistic decreasing from $\xi_2 = 36.4$ to $\xi_2 = 0.57$ on unfolding (cf. 0.05 and 0.01 significance points for $N = 44$ of 7.72 and 10.9 respectively). The fold test therefore is positive at 99% confidence and indicates that the magnetisation found at all sites except US07 was acquired prior to the early Paleozoic Delamerian Orogeny. The tilt corrected mean direction for $N = 44$ specimens is $D = 231.4°$, $I = 41.1°$ ($\alpha_{95} = 5.6°$), yielding a pole position at latitude $\lambda_p = 46.0°$S, longitude $\phi_p = 49.6°$E ($\alpha_{95} = 6.4°$).

Comparison with Australian Ediacaran–Cambrian APWP

The Ulupa pole position is shown in Figure 7, where it is compared with Australia's late Ediacaran–Cambrian apparent polar wander path (APWP). The other poles shown in Figure 7 are listed in Table 2. The APWP spans 150 million years from *ca* 635–490 Ma and places Australia in low paleolatitudes throughout the interval. The poles differ significantly from each other, suggesting Australia underwent continual drift during that time (Schmidt *et al.* 2009) and was rotating about a nearby Euler pole in low paleolatitudes.

The Ulupa pole position is not far from the post-tectonic Black Hill Norite pole. This is significant since the Ulupa magnetisation mostly pre-dates tectonic deformation. This suggests that this orogeny took place during a period of minimal polar wandering, which suggests that it was not a protracted orogeny but occurred during a relatively short period.

AEROMAGNETIC INTERPRETATION

To calculate a representative total magnetisation direction based on the measured properties for the Ulupa Siltstone we have first separated those sites where the remanence is pre-tilting from those where it is primarily post-tilting, the latter being northerly with a moderate to high negative inclination. Most samples with post-tilting magnetisations have directions close to the local geomagnetic field. The mean NRM vector based on samples with pre-tilting remanences, and usually Koenigsberger ratios above unity, is $D = 221.8°$, $I = 38.3°$ (corrected for tilt) and intensity = 1.75 Am^{-1} for $N = 40$. The direction is similar to the mean Characteristic Remanent Magnetisation (ChRM) direction after thermal demagnetisation and identifying linear segments (Table 1). The mean measured Koenigsberger ratio (*Q*) for all 105 specimens is $Q = 1$, calculated by taking the vector mean of the NRMs for each site. If vector means are not used the calculated (scalar) *Q* is 4, which is far too large given that modelling of the airborne magnetic signature shows the effective *Q* is between 1.3 and 1.5.

The Mount Barker syncline, where the Total Magnetic Intensity (TMI) anomaly is clearly dominated by remanent magnetisation, has been modelled using ModelVision^{TM}. Figure 8 shows the TMI image with lows of 58,450 nT and highs of 59 500 nT and a profile oriented NW–SE. The limbs have been modelled using two dipping prisms. The west limb (on the left), dipping 65° E, was modelled with a $Q = 1.3$ ($k = 0.15$ SI) and effective inclination (in the plane perpendicular to the

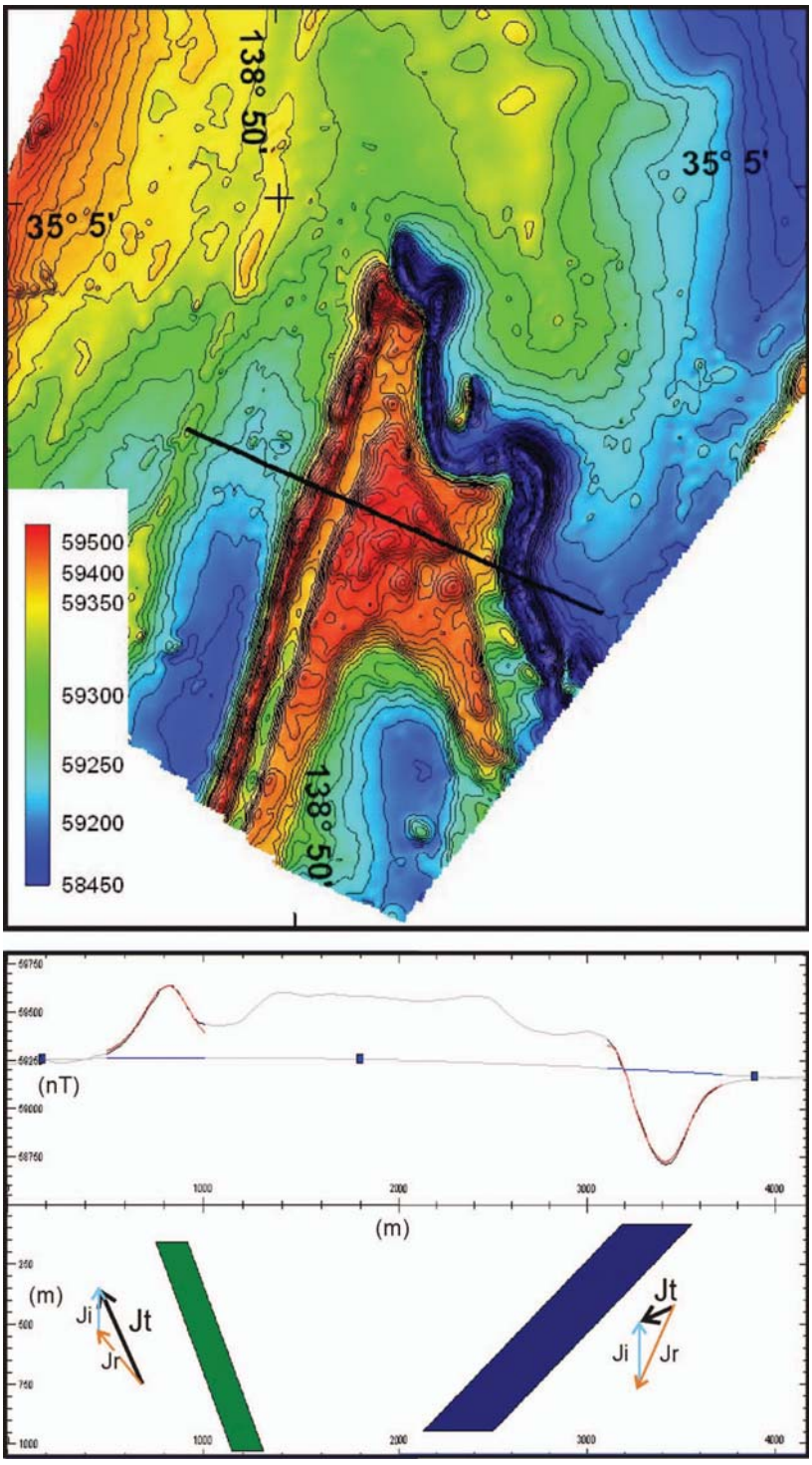


Figure 8 TMI anomaly of a plunging syncline in the Mt Lofty Ranges (south eastern Flinders Ranges) showing the modelled profile, the regional field and remanent (Jr), induced (Ji) and resultant total (Jt) magnetisations. The modelled field is shown in red. The east dipping limb on the left was modelled with a $Q = 1.5$ ($k = 0.025$ SI) and effective inclination (in plane perpendicular to the body) of –65°. The west dipping limb on the right was modelled with a $Q = 1.3$ ($k = 0.029$ SI) and effective inclination (in plane perpendicular to the body) of 16°.

body) of –65°. The east limb (on the right), dipping 40° W, was modelled with a $Q = 1.5$ ($k = 0.06$ SI) and effective inclination (in the plane perpendicular to the body) of +16°. In Figure 8 only the sections of the profiles near centres of the anomalies have been used to constrain the models.

DISCUSSION AND CONCLUSIONS

Remagnetisation of the Ulupa Siltstone in the Mt Lofty Ranges during the Delamerian Orogeny, but mostly prior to deformation, is consistent with the high-temperature, low-pressure nature of the orogeny. While

peak temperatures were attained during the D_2 deformation (Sandiford *et al.* 1995) the conversion of siderite to magnetite would have occurred very early, in the main before deformation began. The Curie temperature found is somewhat higher than that of pure magnetite but this is not unusual for slightly weathered rocks where Fe cations have been leached from the magnetite leaving a slight cation deficiency, i.e. partial maghematisation.

The pole position determined for the Ulupa Siltstone is very similar to that of the post-orogenic Black Hill Norite and although the Ulupa pole suggests some slight back tracking of the pole path, the poles essentially bracket the Delamerian Orogeny indicating little or no polar wandering occurred between the remagnetisation of the Ulupa Siltstone, prior to deformation, and the intrusion/cooling of the Black Hill Norite, after deformation. This brevity is consistent with a short-lived deformation as suggested by the metamorphic mineralogy and isotherm studies of Sandiford *et al.* (1995) in the Mt Lofty Ranges.

Modelling of the aeromagnetic signatures over the Ulupa Formation shows that the magnetisation of the formation is not simply in the direction of the local magnetic field. This is consistent with the measured magnetic properties which indicate that remanent magnetisation is dominant and mostly acquired prior to Delamerian folding.

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APPENDIX 1: SAMPLING SITES

Sites US01–02. Springs Road near Nairn 35°04′57″S, 138°54′34″E (*site PA21* ~160 m east of *site PA01*) (Echunga 1:50 000 Topographic Sheet). Grey micaceous siltstone. Strike 0° (TN), dip 45°E.

Site US03–04. Barker Creek (west limb of Kanmantoo syncline) 35°05′7″S, 138°54′37″E (Euchunga 1:50 000 Topographic Sheet). Strike 0° (TN), dip 45°E.

Site US05. West limb of Macclesfield syncline 35°09′33″S, 138°48′51″E (Euchunga 1:50 000 Topographic Sheet). Grey siltstone. Strike 30° (TN), dip 35°E.

Sites US06. Angaston–Keynton Road, just east of Barossa–west limb of the Moculta anticline 34°31′28″S, 139°04′43″E (Angaston 1:50 000 Topographic Sheet). Grey micaceous siltstone. Strike 337° (TN), dip 70°SW.

Site US07. Angaston–Keynton Road, just east of Barossa–west limb of the Moculta anticline 34°31′17″S, 139°04′33″E (Angaston 1:50 000 Topographic Sheet). Grey micaceous siltstone. Strike 337° (TN), dip 70°SW.

Site US08. West limb of syncline but near closure 34°06′3″S, 139°08′49″E (Angaston 1:50 000 Topographic Sheet). Grey siltstone. Strike 346° (TN), dip 80°E.

Site US09. West limb of syncline but near closure 34°06′46″S, 139°10′39″E (Angaston 1:50 000 Topographic Sheet). Grey siltstone. Strike 346° (TN), dip 75°W.

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